Homework 5

PHYS 6020 - Fall 2023

Due on Friday, September 29th, 2022

Problem 1

As an extension of the problem discussed in class, consider an electron added to the water molecule, which is composed of two Hydrogen and one Oxygen atoms. Because this is a polar molecule, the center of negative charge resides in the Oxygen atom while each of the two identical Hydrogen atoms holds half of the positive charge. We can put this second electron in either of the three sites, thus the Hilbert space of states is spanned by three vectors $|O\rangle; |H_1\rangle$ and $|H_2\rangle$ corresponding to the Oxygen and Hydrogen atoms positions respectively. The energy for the electron to sit on the Oxygen site is $+\hbar\omega_0$ while the energy to sit on either Hydrogen sites is $-\hbar\omega_0$.

(i) Write the matrix corresponding to the potential energy in the basis described above and label this matrix as H_0.

(ii) Because of the underlying polarity, the electron will not 'sit' statically on either of these positions, but move from atom to atom. We define Δ as the kinetic energy of the electron when moving from or to the Oxygen atom from either of the Hydrogen sites, and b when it moves between the two Hydrogen sites. Write the matrix representing the kinetic energy in this situation and name it H_1.

(iii) Do H_0 and H_1 form a C.S.C.O.? Why? why not?

(iv) Find eigenvectors and eigenvalues of $H = H_0 + H_1$.

(v) Sketch the form of the corresponding probability densities in real space associated with each eigenstate. Justify your answer.

Problem 2

Let $\hat{x}$ and $\hat{p}_x$ be the quantum mechanical operators representing position and momentum along the x-direction for a particle moving in one-dimension.

(i) Evaluate the commutator

$$\left[\hat{x}, e^{i\hat{p}_x a/\hbar}\right]$$

(ii) Using the result obtained above, prove that

$$|\beta\rangle = e^{i\hat{p}_x a/\hbar}|x\rangle,$$

where $|x\rangle$ is an eigenstate of the position operator $\hat{x}$, i.e., $\hat{x}|x'\rangle = x'|x'\rangle$.

(iii) What is the eigenvalue associated with this eigenstate?

Problem 3

A box containing a particle is divided into a right and a left compartment by a thin partition. If the particle is known to be on the right (left) side with absolute certainty, the state is represented by the ket $|R\rangle$ (or $|L\rangle$). The most general state vector can then be written as:

$$|\alpha\rangle = |R\rangle\langle R|\alpha\rangle + |L\rangle\langle L|\alpha\rangle$$

where $\langle R|\alpha\rangle$ and $\langle L|\alpha\rangle$ can be regarded as 'wave-functions' in this basis. The particle can tunnel through the partition, this tunneling process is characterized by a Hamiltonian:

$$H = \Delta(|L\rangle\langle R| + |R\rangle\langle L|)$$

where Δ is a number with dimensions of energy.

(i) Find the normalized energy eigenvectors. What are the corresponding energy eigenvalues?

(ii) In the Schrödinger picture, the basis kets $|R\rangle$ and $|L\rangle$ are fixed, and the state vector moves in time. Suppose the system is represented by the ket $|\alpha\rangle$ as given above at $t = 0$. Find the state vector $|\alpha, t_0 = 0; t\rangle$ for $t > 0$ by applying the appropriate time evolution operator to $|\alpha\rangle$.

(iii) Suppose that at $t = 0$ the particle is on the right side with certainty. What is the probability for observing the particle on the left side as a function of time?

(iv) Write down the coupled Schrödinger equations for the wave functions $\langle R|\alpha, t_0 = 0; t\rangle$ and $\langle L|\alpha, t_0 = 0; t\rangle$. Show that the solutions to the coupled equations are what you expect from (ii).

(v) Suppose the printer made an error and wrote H as:

$$H = \Delta|L\rangle\langle R|$$

By explicitly solving the most general time-evolution problem with this Hamiltonian, show that the probability conservation is violated. Why?

Problem 4

Solve Problem 3.6 at the end of Chapter 3 of Zettili's book. Is this wavefunction normalized? Do you need it to be to answer the questions posed?

Problem 5

Solve Problem 3.7 at the end of Chapter 3 of Zettili's book.